FINANCE TRACKER

Initial Project Requirements Document

Course: Software Construction and Development (SCD) Lab
Development Approach: Agile and Incremental Development
Programming Language: Python
Repository Link: https://github.com/SuhaibAhmad0507/FinanceTracker

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1. Group Members

No.	Name	Roll No	GitHub Username	Initial Responsibility
1	Suhaib Ahmad	42	SuhaibAhmad0507	Not decided yet
2	Muhammad Rehan Khan	31	DevRehan5599	Not decided yet
3	Maaz Khan	21	Maaz-Khann	Not decided yet

2. Project Description

The Finance Tracker is a Python-based application intended to help users record income and expenses, categorize financial transactions, review their transaction history, monitor their available balance, and manage spending limits.

The application will be developed incrementally. The team will maintain the source code, requirements, and development history in a shared public GitHub repository.

3. Problem Statement

People who track their finances manually may find it difficult to maintain accurate transaction records, determine how much money remains available, identify spending patterns, and stay within their budgets.

The Finance Tracker aims to address these problems by providing a simple application for recording and reviewing personal financial information.

4. Project Objectives

The objectives of this project are to:

1. Provide a way to record income and expenses.
2. Organize financial transactions into meaningful categories.
3. Allow users to view and search previous transactions.
4. Calculate the current balance and summarize spending.

5. Allow users to define budgets and identify when spending exceeds the specified limits.
6. Apply collaborative software development practices through Git and GitHub.

5. Initial Functional Requirements

Requirement ID	Description	Priority
FR-01	The system shall allow users to add income transactions with relevant details such as amount, date, and description.	High
FR-02	The system shall allow users to add expense transactions with relevant details such as amount, date, and description.	High
FR-03	The system shall allow transactions to be assigned categories.	High
FR-04	The system shall display transaction history and allow users to search for transactions.	Medium
FR-05	The system shall calculate the current balance based on recorded income and expenses.	High
FR-06	The system shall display spending summaries to help users understand their expenses.	Medium
FR-07	The system shall allow users to set budget limits and alert them when spending exceeds the applicable limit.	Medium

These requirements describe the initial scope and may be refined after the team discusses implementation details.

6. Non-Functional Requirements

6.1 Usability

The application should have a simple and understandable interface suitable for users with basic computer skills.

6.2 Reliability

The system should validate transaction inputs and handle invalid values without terminating unexpectedly.

6.3 Data Integrity

Financial calculations should be accurate, and transaction records should be stored consistently.

6.4 Maintainability

The source code should be organized into clear functions or modules, with meaningful names and appropriate comments where necessary.

6.5 Portability

The application should run on systems with a compatible Python installation and any required dependencies.

7. Assumptions

- Users will enter their financial transactions manually.
- Users will provide valid dates, descriptions, and transaction amounts.
- The application will initially focus on a single user's personal finances.
- The team will select an appropriate storage method during development.
- The application will use Python as its primary programming language.

8. Constraints and Known Limitations

- The project is being developed within the time and resource limits of a university semester.
- The initial version will not connect directly to bank accounts or payment services.
- Automatic bank transaction imports and cloud synchronization are not part of the initial scope.
- Authentication and multi-user support are not confirmed requirements for the initial version.
- The choice of storage technology and user interface remains subject to the team's implementation decisions.
- The application's financial summaries will depend on the completeness and accuracy of the information entered by users.

10. Technology and Development Tools

Tool or Technology	Intended Use
Python	Application development
Git	Source control and change history
GitHub	Shared repository and collaboration

The team may select additional tools if they are needed during development.